Automatic Shade System



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **Medium** 2 labs

Covers these lab items: Automatic shade system

What it is used for

An automatic shade system adjusts shade screens or coverings to control sunlight, temperature, and light exposure in a greenhouse or growing area. Correct setup and sensor control help protect crops and maintain suitable growing conditions.

Main parts

- Shade cloth, screen, or covering material
- Support frame, rails, and mounting brackets
- Motor or actuator and drive mechanism
- Light, temperature, or weather sensors if fitted
- Controller, timer, and limit switches
- Power supply and manual override control

Before you start

1. Read the installation and operating instructions for the specific shade system.
2. Inspect the shade cloth, frame, rails, cables, motor, and mounting points for damage.
3. Confirm that the structure is secure and that the shade can move without obstruction.
4. Check sensor placement and verify that the controller settings suit the crop and growing conditions.
5. Test the emergency stop or manual override if fitted and confirm the movement limits are set.
6. Ensure the power supply and protective devices meet the manufacturer's requirements.

How to use

1. Switch on the controller and select the approved automatic or manual operating mode.
2. Set the required light, temperature, timer, or weather thresholds according to the crop plan.
3. Run a brief movement test to confirm that the shade opens and closes smoothly.
4. Observe the sensors and controller response as environmental conditions change or during a controlled test.
5. Adjust the threshold settings only within the approved growing plan and equipment limits.
6. Check that the shade stops at the end positions without excessive tension or mechanical impact.
7. Inspect the covered area for uniform shading and verify that crops receive the intended light level.
8. Use the manual override only when required and return the system to the approved mode afterward.

9. Stop the system if movement becomes uneven, the frame shifts, or a motor or cable shows a fault.
10. Record settings, test results, and maintenance needs according to the facility procedure.

⚠ Safety

- Keep hands and tools clear of moving screens, rails, cables, and actuator linkages.
- Isolate power before cleaning, adjusting, or repairing the drive mechanism.
- Do not climb on the shade structure or use it as a support platform.
- Keep people away from the movement path while the system operates.
- Stop operation during unsafe wind or weather conditions specified by the manufacturer.
- Do not bypass limit switches, guards, or other fitted safety devices.

Care & storage

- Remove dust, plant debris, and other material that obstructs shade movement.
- Inspect cloth, stitching, rails, fasteners, motors, cables, and sensors regularly.
- Keep sensors clean and verify their readings according to the maintenance schedule.
- Arrange servicing of actuators and controllers according to the manufacturer's instructions.

Common mistakes to avoid

- Incorrect sensor thresholds can expose crops to excessive heat or insufficient light.
- Obstructions in the rails can damage the shade cloth or motor.
- Ignoring wind limits can damage the screen and supporting structure.
- Leaving manual override active can prevent automatic adjustment from working as intended.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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